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① Internet Protocols and HTTP

a) TCP/IP, DNS and URL

- TCP/IP: Transfers data b/w browser and server
- DNS: converts domain names into IP addresses
- URL: specifies the web resources address

b) HTTP messages

- Request line: method, URL, version
- Headers: additional information.
- Body: submitted data.
- Status line: Response status.
- Response body: returned content

(39) ✓

c) HTTP methods

method	Purpose	Safe	idempotent
Get	Read	Yes	Yes
Post	submit	NO	NO
Put	Update	NO	YES
Delete	Delete	NO	Yes

d) Status codes

- 200 - success.
- 301 - Permanent redirect
- 304 - Not modified
- 401 - unauthorized
- 404 - Not found
- 500 - Server error
- 503 - Service unavailable

e) HTTP cookies and HTTPS

- HTTP is stateless.
- cookies store browser information
- session maintain login information
- HTTPS/TLS encrypts communication

2) XHTML course Registration

a) Document structure

- DOCTYPE : Defines document type
- html : Root element
- head : metadata and title
- body: visible content
- header, nav, main, section, footer : Semantic elements.

b) Table elements

- caption : Table title
- thead : Header
- tbody : Data
- th: Heading
- scope : Identifies header.
- colspan: combines columns
- rowspan: combines rows.

c) From elements

- form: Creates form
- action: Submission URL
- method: GET (or) POST
- label : Field name
- input : Accepts data
- select : Dropdown
- required: mandatory field
- Place holder: Hint.

d) Links and images

- <a> creates links
- <img> displays images
- Relative URL: images/1090.png
- Absolute URL: https://example.com/1090.png

HTML vs XHTML

HTML	XHTML
flexible	strict
closing tags sometimes optional	Every tag closed
case-in sensitive	Lowercase
flexible nesting	proper nesting
Attributes may be unquoted	Quotedattributes

* XHTML.

```
xml
<br/>
<input type="text" />
```

3) xml and css

a) xml catalogue.

xml Example

```
<?xml version="1.0"?>
<catalogue xmlns="http://example.com/books">
<book ISBN="101">
<title>HTML</title>
<author>John</author>
<Publisher>ABC</Publisher>
<Year>2024</Year>
<Price>400</Price>
</book>
```

DTD:-

```
<!Element catalogue (book+)>
<!Element book (title ;author , Publisher, Year , Price)>
<! ATTList book ISBN CDATA #Required>
<!element title (#PCData)>
<! element author (#PCData)>
<! Element price (#PCData)>
```

- well-formed xml : follows xml syntax.
- valid xml : follows DTD or xml schema

* Three errors:
1. missing closing tags
2. Improper nesting
3. unquoted attributes.

* XML Vs HTML vs JSON
- XML : Stores and transfers data
- HTML : Displays web Pages
- JSON : Light weight data exchange.

XSLT : converts XML into HTML

CSS : styles XML content.

3b) CSS cascade

Given colours:
- element : Blue
- class : Green
- ID : Red
- Inline : orange

* CSS Types
1. Inline - Inside an element
2. Internal - Inside <style>.
3. External - Separate css file

* Selector

Selector	Example
element	p
class	.box
ID	#header
Descendant	div p
child	div > p
Pseudo-class	a:hover

4) Responsive web Design

a) Box model

- content : actual content
- Padding : Inner space
- Border : element boundary
- margin : outer space
- box - sizing : border - box : includes Padding and border in width
- margin collapsing : Adjacent Vertical margins combine.

b) Positioning

Position	meaning
static	Default
Relative	Relative to normal position
absolute	Relative to Positioned Parent
fixed	fixed on screen
sticky	sticks during scrolling

c) flexbox and Grid

- flexbox one - dimensional layout
- Grid : Two - dimensional layout
- flex box is suitable for navigation
- Grid is suitable for page layout

d) Responsive Techiques

- %, rem, vw : Relative units
- view port : supports mobile scaling
- media queries : Adapt screen layouts
- flexible images : Prevent overflow

e) Typography
- Use readable fonts
- maintain colour contrast
- We CSS variables for consistent styling

5) Java script and online Book store

a) Variable and data Types
- let - changeable variable
- const - cannot be reassigned
- Array - store multiple values
- object - stores key-values pair.

```
let age = 21
const price = 500,
let student = {
  name: "Rasool"
  age: 21
};
const add (a,b) => a+b;
```

6) Dom multiplication:
Dom allows Java script to modify HTML
- get element Bild () - selects ID
- query selector () - select elements
- create element () - creates elements
- class list - changes CSS classes.

c) Event Handling:
click - Button click
Input - Data entry
Event delegation - Parent handles children

d) Regular expression
- Name : ^[A-Za-z]+$
- Email : ^[^\s@]+@[^\s@]+\.[^\s@]+$
- Pass: minimum 8characters one uppercase one digit